Global sensitivity analysis of a supercritical extraction model

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ABSTRACT

This study investigates the process of chamomile oil extraction from flowers. A parameter-distributed model consisting of a set of partial differential equations is used to describe the governing mass transfer phenomena in a cylindrical packed bed with solid chamomile particles under supercritical conditions using carbon dioxide as a solvent. The concept of quasi-one-dimensional flow is applied to reduce the number of spatial dimensions. The flow is assumed to be uniform across any cross-section, although the area available for the fluid phase can vary along the extractor. The physical properties of the solvent are estimated from the Peng-Robinson equation of state. The empirical correlations used in the model are based on laboratory experiments performed under multiple constant operating conditions: $30-40$ °C, $100-200$ bar and $3.33-6.67\cdot10^{-5}$ kg/s. The first and higher-order Sobol indices were computed to assess separate and joined contribution of the operating conditions to the model output and its variance.

1. Introduction

Supercritical CO2 extraction is a green extraction method that uses carbon dioxide in a supercritical state (above its critical point of 31.1 °C and 73.8 bar) to extract compounds from materials. This is one of the most popular applications of supercritical fluids, which was described by many researchers, for example Sodeifian and Sajadian [1], Reverchon et al. [2] and Sovova [3]. Traditional methods, such as distillation and organic solvent extraction, are commonly employed although they have drawbacks. Distillation involves high temperatures that can lead to the thermal degradation of heat-sensitive compounds. This limitation has increased the popularity of alternative techniques, such as supercritical fluid extraction. Supercritical CO_2 is attractive due to its distinctive properties: it is inflammable, non-toxic and non-corrosive. Supercritical fluids can exhibit both gas- and liquid-like properties, allowing for adjustable dissolving power through changes in operating conditions.

Applications for supercritical carbon dioxide are not limited to extraction processes; they can also be used for impregnation, as described by Weidner [4] and Machado et al. [5]. Impregnation is defined as modification of the properties of bulk substances by physically or chemically binding/adsorbing impregnates to a bulk material or surface, for example, the surface hydrophobisation. The main advantage of using supercritical CO_2 is that, after depressurisation, it desorbs from the surface and evaporates, leaving a solvent-free product. However, the main disadvantage of using carbon dioxide for impregnation is the low solubility of many drugs surface hydrophobisation. The study of Ameri et al. [6] investigates the loading of lansoprazole into polymers using supercritical carbon dioxide and examines how various parameters, such as temperature, pressure and time, affect the drug loading efficiency. The results indicate that increasing any of these parameters enhances drug loading, with temperature having the most significant impact. Fathi et al. [7] explored the use of supercritical carbon dioxide to enhance the bioavailability of ketoconazole by impregnation into water-soluble polymers, specifically polyvinylpyrrolidone and hydroxypropyl methylcellulose. Utilising a Box-Behnken design, the researchers optimised the impregnation process by varying pressure, temperature and time, achieving an increased drug loading range.

Another application of supercritical CO_2 is nanoparticle formation, as investigated by Padrela et al. [8], Franco and De Marco [9] and Sodeifian et al. [10]. Supercritical CO_2-assisted technologies enable the production of different morphologies of different sizes, including nanoparticles and nanocrystals, by modulating the operating conditions. Supercritical fluid-based processes have advantages over techniques conventionally employed to produce nano-sized particles or crystals, such as reduced use of toxic solvents. Moreover, the CO_2 can be removed from the final product by simple depressurisation. Sodeifian and Sajadian [11] investigated the solubility of Letrozole, a poorly water-soluble anticancer drug, in supercritical carbon dioxide with and without menthol as a solid co-solvent. The addition of menthol increased the solubility of Letrozole by 7.1 times compared to supercritical CO_2 alone. Using the rapid expansion of supercritical solutions with the solid co-solvent method, the average particle size of Letrozole was reduced to the nanoscale. Temperature was found to have the most significant impact on nanoparticle size reduction, while pressure had the least effect. The study of S. Ardestani et al. [12] explored the preparation of phthalocyanine green nano pigment using supercritical carbon dioxide as an anti-solvent. The researchers employed the gas anti-solvent technique to achieve nano-sized pigment particles. Sodeifian and Sajadian [13] analysed the production of amiodarone hydrochloride nanoparticles using ultrasonic-assisted rapid expansion of the supercritical solution in a liquid solvent method. The researchers achieved significant particle size reduction by optimising parameters such as pressure, temperature and

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polymeric stabiliser concentration. Characterisation techniques confirmed the successful formation of nanoparticles with improved properties.

Although the processes discussed are based on different phenomena, the outcome of these technologies strongly depends on the solubility in supercritical CO_2. The solubility of the solid in the solvent can be important for determining the partition coefficient for a potential impregnation and whether a cosolvent should be used in a supercritical extraction of a compound. Moreover, a good solute solubility not only accelerates the initial stages of the extraction but also reduces the time of the extraction process. Multiple researchers, such as Bagheri et al. [14], Tabebordbar et al. [15], Sheikhi-Kouhsar et al. [16] or Bagheri et al. [17], have conducted extensive research to measure the solubility of different solutes in the supercritical CO_2 and on the development of mathematical models. Several methods, including density-based correlations and equation of state, are used for correlations of solute solubility in the supercritical CO_2. The semi-empirical models are often used due to their ease of use and the absence of the need for physicochemical properties like sublimation pressure and critical features, which cannot be directly measured experimentally.

The present study investigates the extraction of essential oil from chamomile flowers (*Matricaria chamomilla L.*) via supercritical fluid extraction techniques in addition to the modelling of this process. Chamomile is a medicinal herb widely cultivated in southern and eastern Europe - in countries such as Germany, Hungary, France and Russia. It can also be found outside Europe, for instance in Brazil, as discussed by Singh et al. [18]. Chamomile is distinguished by its hollow, bright gold cones that house disc or tubular florets surrounded by about fifteen white ray or ligulate florets. The plant has been utilised for its medicinal benefits, serving as an anti-inflammatory, antioxidant, mild astringent, and healing remedy. Chamomile extracts are widely used to calm nerves and mitigate anxiety, hysteria, nightmares, insomnia and other sleep-related conditions, according to Srivastava [19]. Orav et al. [20] reported that oil yields from dried chamomile samples ranged from 0.7 to 6.7 mL/kg. The highest yields of essential oil, between 6.1 mL/kg and 6.7 mL/kg, were derived from chamomile sourced from Latvia and Ukraine. In comparison, chamomile from Armenia exhibited a lower oil content of 0.7 mL/kg. Milovanovic and Grzegorczyk [21] extracted oils from Roman chamomile seeds in a two-stage process. First, supercritical carbon dioxide at pressures of up to 450 bar and temperatures up to 60 °C was used for the first extraction. Then, the oil was re-extracted using supercritical CO_2 with the addition of ethanol. Through optimisation of the operating pressure, temperature, production cost, fraction of milled seeds and addition of co-solvent, the amount of separated chamomile oil increased from 2.4 to 18.6% and the content of unsaturated fatty acids up to 88.7%.

The literature offers various mathematical models to describe the extraction of valuable compounds from biomass. Selecting a process model is case-dependent and requires analysis of each model's specific assumptions about mass transfer and thermodynamic equilibrium. Depending on the case, two approaches can be considered when developing a mathematical model for the extraction process. A model based on multiple regression can be used if the relation between inputs and outputs is the only one of interest. Sodeifian et al. [22] investigated the influence of pressure, temperature and particle size on the extraction efficiency of oil from *Dracocephalum kotschyi Boiss* seed. A second-order polynomial model was applied to obtain the corresponding response surface and to identify the optimum operating conditions. The study of Sodeifian et al. [23] investigates the extraction of essential oil from Eryngium billardieri, focusing on optimising the extraction conditions and developing a mathematical model based on the second-order polynomial to predict the process yield. The researchers employed a simulated annealing algorithm to optimise parameters such as pressure, temperature and extraction time, aiming to maximise oil extraction efficiency.

Alternatively, a first-principle model can be derived and applied to cover not only the input-output relations, but also the phenomena occurring in the system. This approach allows for a more detailed representation of the system behaviour, but requires deeper understanding of the underlying physics and more rigorous experiments. Sliczniuk and Oinas [24] provides a list of studies on SFE, including modelling approaches, raw materials, and operating conditions. Below some of the most popular SFE models are discussed.

Goto et al. [25] presented the shrinking core (SC) model, which describes a process of irreversible desorption followed by diffusion through the pores of a porous solid. When the mass transfer rate of the solute in the non-extracted inner region is significantly slower than in the outer region, where most of the solute has already been extracted or when the solute concentration exceeds its solubility in the solvent, a distinct boundary may form between the inner and outer regions. As extraction progresses, the core of the inner region shrinks. The model envisions supercritical CO_2 extraction as a sharp, inward-moving front, with a completely non-extracted core ahead of the front and a fully extracted shell behind it.

Sovova [3] proposed the broken-and-intact cell (BIC) model, which assumes that a portion of the solute, initially stored within plant structures and protected by cell walls, is released during the mechanical breakdown of the material. The solute located in the region of broken cells near the particle surface is directly exposed to the solvent, while the core of the particle contains intact cells with undamaged walls. This model describes three extraction phases: a fast extraction phase for accessible oil, a transient phase and a slow phase controlled by diffusion. The model has been successfully applied to the extraction of grape oil (Sovová et al. [26]) and caraway oil (Sovova et al. [27]).

The supercritical fluid extraction (SFE) process can be treated similarly to heat transfer, considering solid particles as spheres cooling down in a uniform environment. Bartle et al. [28] introduced the hot ball diffusion (HBD) model, where spherical particles with a uniformly distributed solute diffuse similarly to heat diffusion. Unlike the BIC model, where the solute is readily available on the particle surface, the HBD model is suitable for systems with small quantities of extractable materials and is not limited by solubility. The model is particularly relevant when mass transfer is controlled by internal diffusion, allowing results from single particles to be extended to the entire bed under uniform conditions. Reverchon et al. [2] have further elaborated on the HBD model and used it to simulate extraction processes.

Reverchon [29] proposed a model for the extraction of essential oils that are mainly located inside vegetable cells in organelles called vacuoles. Only a small fraction of essential oil might be near the particle surface due to the breaking up of cells during grinding or in epidermal hairs located on the leaf surface. The fraction of oil freely available on the particle surface should not be significant in the case of SFE from leaves. Consequently, SFE of essential oil from leaves should be controlled mainly by internal mass-transfer resistance. Therefore, the external mass-transfer coefficient was neglected in the development of Reverchon's model (Reverchon [29]). The mass balances were developed with the additional hypotheses that axial dispersion can be neglected and that the solvent density and flow rate are constant along the bed.

This work builds upon the linear kinetic model suggested by Reverchon [29], deriving fundamental governing equations to develop a comprehensive model for the chamomile oil extraction process. The model aims at control-oriented simplicity, assuming semi-continuous operation within a cylindrical vessel. The process involves a supercritical solvent being pumped through a fixed bed of finely chopped biomass to extract the solute, followed by separation of the solvent and solute in a flush drum to collect the extract. Parameters such as pressure (P), feed flow rate (F) and inlet temperature (T_{in}) are adjustable and measurable, while the outlet temperature (T_{out}) and the amount of product at the outlet can only be monitored. Figure 1 presents a simplified process flow diagram.

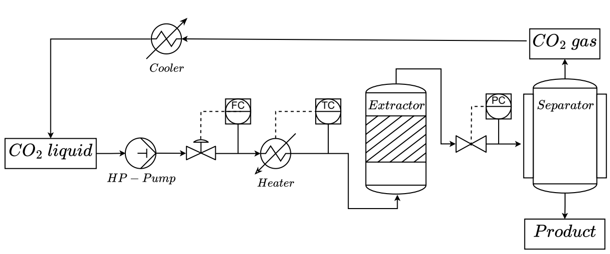

Figure 1: Process flow diagram.

One limitation of the model development process is the need to capture the required physical behaviour while maintaining interpretability. However, models inherently contain a large number of parameters, creating a complex network of dependencies. Model analysis techniques and sensitivity analysis provide theoretical frameworks to identify dependencies between model output and its parameters. As discussed by many authors such as Jamett et al. [46], Duong et al. [47] or Lucay et al. [48], sensitivity analysis (SA) is the study of uncertainties in the responses (outputs) of a mathematical model, which can be attributed to different sources of uncertainty in the values of the model parameters (input variables). These uncertainties can be epistemic (related to lack of knowledge) or stochastic (due to randomness). The effects of these uncertainties must be examined to avoid poor design, safety issues, and unreliable systems, among other problems. SA methods can help to analyse the effect of uncertainties. However, they can be computationally expensive for complex issues of chemical processes.

As noted by Gan et al. [49], the literature presents several methods for performing SA. In general, these methods can be classified into two main groups: local SA and global SA (GSA). The local SA explores changes in the model response by varying one parameter while keeping other parameters constant. In contrast, the GSA examines changes in the model response by simultaneously varying all parameters.

2. Materials and methods

2.1. Governing equations

The governing equations for a quasi-one-dimensional flow were derived by following the work of Anderson [39]. A quasi-one-dimensional flow refers to a fluid flow scenario which assumes that the flow properties are uniformly distributed across any cross-section. This simplification is typically applied when the cross-sectional area of the flow channel changes, such as through irregular shapes or partial filling of an extractor. According to this assumption, velocity and other flow properties change solely in the flow direction.

As discussed by Anderson [40], all flows are compressible, but some can be treated as incompressible since the velocities are low. This assumption leads to the incompressible condition: $\nabla \cdot u = 0$, which is valid for constant density (strictly incompressible) or varying density flow. The assumption allows for removal of acoustic waves and large perturbations in density and/or temperature. In the 1-D case, the incompressibility condition becomes $\frac{du}{dz} = 0$, so the fluid velocity is constant along $z-$ direction.

The set of quasi-one-dimensional governing equations in Cartesian coordinates is described by Equations 1 - 3:

$$\frac{\partial\left(\rho_f A_f\right)}{\partial t} + \frac{\partial\left(\rho_f A_f v\right)}{\partial z} = 0 \tag{1}$$

$$\frac{\partial\left(\rho_f v A_f\right)}{\partial t} + \frac{\partial\left(\rho_f A_f v^2\right)}{\partial z} = -A_f \frac{\partial P}{\partial z} \tag{2}$$

$$\frac{\partial\left(\rho_f e A_f\right)}{\partial t} + \frac{\partial\left(\rho_f A_f v e\right)}{\partial z} = -P\frac{\left(A_f v\right)}{\partial z} + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) \tag{3}$$

where ρ_f is the fluid density, A_f is the function which describes a change in the cross-section, v is the velocity, P is the total pressure, e is the internal energy of the fluid, T is the temperature, t is time, and z is the spatial direction.

Equation 1-3 are the quasi-one-dimensional conservation laws for a variable-area duct representing a packed bed under the quasi-1D and low-Mach assumptions. Equation (1) (continuity) enforces conservation of mass is convected axially with velocity. Equation (2) (axial momentum) balances the momentum of the fluid parcel with the pressure gradient. The viscous wall/porous drag and body forces are neglected in the control-oriented model because pressure is imposed and nearly uniform in the low-velocity regime. Equation (3) (energy) transports internal energy by advection, includes pressure work, and accounts for axial heat conduction via the effective conductivity. The details on treatment of the governing equations are presented in Section 2.2.

2.2. Extraction model

2.2.1. Continuity equation

The previously derived quasi-one-dimensional continuity equation (Equation 1) is redefined by incorporating the function $A_f = A\phi$. This modification differentiates constant and varying terms, where the varying term accounts for changes in the cross-sectional area available for the fluid. Equation 4 shows the modified continuity equation:

$$\frac{\partial(\rho_f \phi)}{\partial t} + \frac{\partial(\rho_f v A \phi)}{\partial z} = 0 \tag{4}$$

where A is the total cross-section of the extractor and ϕ describes the porosity along the extractor.

Assuming that the porosity and mass flow rate are constant in time, the temporal derivative becomes the mass flux F, and the spatial derivative can be integrated along z as

$$\int \frac{\partial(\rho_f v A \phi)}{\partial z} dz = F \rightarrow F = \rho_f v A \phi \tag{5}$$

To simplify the system dynamics, it is assumed that F is a control variable and affects the whole system instantaneously (due to $\nabla \cdot u = 0$), which allows the velocity profile that satisfies mass continuity based on F, ϕ and ρ_f to be found:

$$v = \frac{F}{\rho_f A \phi} \tag{6}$$

Similarly, superficial velocity may be introduced:

$$u = v\phi = \frac{F}{\rho_f A} \tag{7}$$

The fluid density ρ_f can be obtained from the Peng-Robinson equation of state if the temperature and thermodynamic pressure are known along z. Variation in fluid density may occur due to pressure or inlet temperature changes. In a non-isothermal case, in Equations 6 and 7 ρ_f is considered to be the average fluid density along the extraction column.

2.2.2. Mass balance for the fluid phase

Equation 8 describes the movement of the solute in the system, which is constrained to the axial direction due to the quasi-one-dimensional assumption. Given that the solute concentration in the solvent is negligible, the fluid phase is described as pseudo-homogeneous, with properties identical to those of the solvent itself. It is also assumed that the thermodynamic pressure remains constant throughout the device. The analysis further simplifies the flow dynamics by disregarding the boundary layer near the extractor's inner wall. This leads to a uniform velocity profile across any cross-section perpendicular to the axial direction. Thus, the mass balance equation includes convection, diffusion and kinetic terms representing the fluid phase behaviour:

$$\frac{\partial c_f}{\partial t} + \frac{1}{\phi}\frac{\partial\left(c_f u\right)}{\partial z} = \frac{1-\phi}{\phi} r_e + \frac{1}{\phi}\frac{\partial}{\partial z}\left(D_e^M \frac{\partial c_f}{\partial z}\right) \tag{8}$$

where c_f represents the solute concentration in the fluid phase, r_e is the mass transfer kinetic term and D_e^M is the axial dispersion coefficient.

2.2.3. Mass balance for the solid phase

As given by Equation 9, the solid phase is considered to be stationary, without convection and diffusion terms in the mass balance equation. Therefore, the only significant term in this equation is the kinetic term of Equation 10, which connects the solid and fluid phases. For simplicity, the extract is represented by a single pseudo-component:

$$\frac{\partial c_s}{\partial t} = \underbrace{-r_e}_{\text{Kinetics}} \tag{9}$$

2.2.4. Kinetic term

As the solvent flows through the fixed bed, CO_2 molecules diffuse into the pores, adsorb on the inner surface and form a film due to solvent-solid matrix interactions. The dissolved solute diffuses from the particle's core through the solid-fluid interface, the pore and the film into the bulk. Figure 2 shows the mass transfer mechanism, where the mean solute concentration in the solid phase is denoted as c_s, and the equilibrium concentrations at the solid-fluid interface are denoted as c_s^* and c_p^* for the solid and fluid phases, respectively. The concentration of the solutes in the fluid phase in the centre of the pore is denoted as c_p. As the solute diffuses through the pore, its concentration changes, reaching c_{pf} at the opening. Then, the solute diffuses through the film around the particle and reaches bulk concentration c_f. The two-film theory describes the solid-fluid interface inside the pore. The overall mass transfer coefficient can be determined from the relationship between the solute concentration in one phase and its equilibrium concentration.

Bulley et al. [41] suggest a process where the driving force for extraction is given by the difference between the concentration of the solute in the bulk, c_f, and in the centre of the pore, c_p^*. According to the equilibrium relationship, the concentration c_p^* is in equilibrium with c_s. The rate of extraction is thus $r_e\left(c_f - c_p^*(c_s)\right)$. In contrast, Reverchon [29] proposes a driving force given by the difference between c_s and c_p^*. As given by Equation 10, the concentration c_p^* is determined by the equilibrium relationship with c_f:

$$r_e = \frac{D_i}{\mu l^2}\left(c_s - c_p^*\right) \tag{10}$$

where μ is sphericity, l a characteristic dimension of particles that can be defined as $l = r/3$, r is the mean particle radius, D_i corresponds to the overall diffusion coefficient

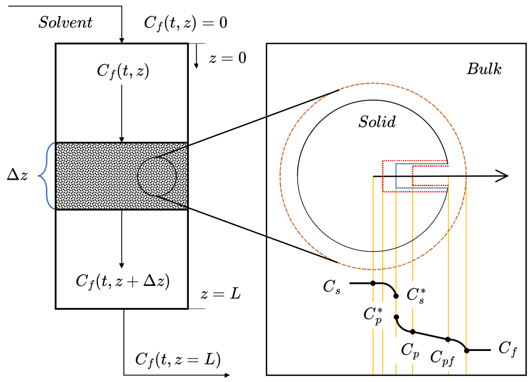

Figure 2: Mass transfer mechanism.

and c_P^* is the concentration at the solid-fluid interface (which according to the internal resistance model is supposed to be at equilibrium with the fluid phase).

According to Bulley et al. [41], a linear equilibrium relationship (Equation 11) can be used to find the equilibrium concentration of the solute in the fluid phase c_f^* based on the concentration of the solute in the solid phase c_s:

$$c_f^* = k_p c_s \tag{11}$$

The volumetric partition coefficient k_p acts as an equilibrium constant between the solute concentration in one phase and the corresponding equilibrium concentration at the solid-fluid interphase. Spiro and Kandiah [42] proposed a definition of the mass partition coefficient k_m and the solid density ρ_s as

$$k_m = \frac{k_p \rho_s}{\rho_f} \tag{12}$$

According to Reverchon [29], the kinetic term becomes

$$r_e = \frac{D_i}{\mu l^2}\left(c_s - \frac{\rho_s c_f}{k_m \rho_f}\right) \tag{13}$$

2.2.5. Uneven solute distribution in the solid phase

Following the idea of the broken-and-intact cell (BIC) model (Sovova [43]), the internal diffusion coefficient D_i is considered to be a product of the reference value of D_i^R and the exponential decay function γ, as given by Equation 14:

$$D_i = D_i^R \gamma(c_s) = D_i^R \exp\left(-\Upsilon\left(1 - \frac{c_s}{c_{s0}}\right)\right) \tag{14}$$

where Υ describes the curvature of the decay function. Equation 15 describes the final form of the kinetic term:

$$r_e = \frac{D_i^R \gamma}{\mu l^2}\left(c_s - \frac{\rho_s c_f}{k_m \rho_f}\right) \tag{15}$$

The γ function limits the solute's availability in the solid phase. Similarly to the BIC model, the solute is assumed to be contained inside the cells, some of which are open because the cell walls have been broken by grinding, with the rest remaining intact. The diffusion of solute from a particle's core takes more time than the diffusion of solute close to the outer surface. The same idea can be represented by the decaying internal diffusion coefficient, where the decreasing term is a function of the solute concentration in the solid.

An alternative interpretation of the decay function γ involves taking into account the porous structure of the solid particles, where the pores are initially saturated with the solute. During extraction, the solute within these pores gradually dissolves into the surrounding fluid. Initially, the solute molecules near the pore openings dissolve and diffuse rapidly due to the short diffusion paths. As extraction progresses, the dissolution front moves deeper into the pore structure and solute from the inner regions of the pores begins to dissolve. The diffusion of solute molecules from the interior of the pores to the external fluid becomes progressively slower because the effective diffusion path length increases. This lengthening of the diffusion path enhances mass transfer resistance, reducing the overall diffusion rate.

In an extreme case, this model could be compared with the shrinking core (SC) model presented by Goto et al. [25], where the particle radius is reduced as the solute content in the solid phase decreases. In the SC model, the reduction in particle size leads to significant changes in both the diffusion path length and the surface area available for mass transfer. The diminishing particle size increases the diffusion path within the remaining solid core and decreases the external surface area, both of which contribute to a slower extraction rate. When comparing this to the varying diffusion coefficient in our model, some conceptual similarities can be noticed.

2.2.6. Empirical correlations

The empirical correlations for D_i and Υ were derived by Sliczniuk and Oinas [44] and validated for temperatures of $30-40\,°C$, pressures of $100-200$ bar and mass flow rates of $3.33 - 6.67 \cdot 10^{-5}$ kg/s. Figures 3 and 4 show the results of multiple linear regression applied to parameter estimation solutions and selected independent variables. The region marked with the white dashed line represents the confidence region where the model has been tested. Both correlations should be equal or greater than zero to avoid unphysical behaviour such as reverse mass transfer. The multiple linear regression functions are combined with the rectifier function to ensure non-negativity.

2.2.7. Heat balance

The heat balance equation describes the evolution of enthalpy in the system, given by Equation 16:

$$\frac{\partial\left(\rho_f h A_f\right)}{\partial t} = -\frac{\partial\left(\rho_f h v\right)}{\partial z} + \frac{\partial\left(P A_f\right)}{\partial t} + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) \tag{16}$$

If the value of enthalpy h is known from the time evolution of the energy equation and pressure P is known from measurement, then the temperature T can be reconstructed based on the departure function. The departure function is a mathematical function that characterises the deviation of a thermodynamic property (enthalpy, entropy or internal energy) of a real substance from that of an ideal gas at the

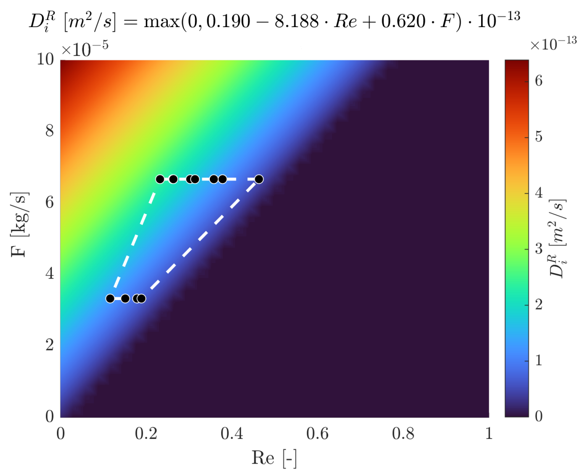

Figure 3: Multiple linear regression $D_i^R = f(Re, F)$.

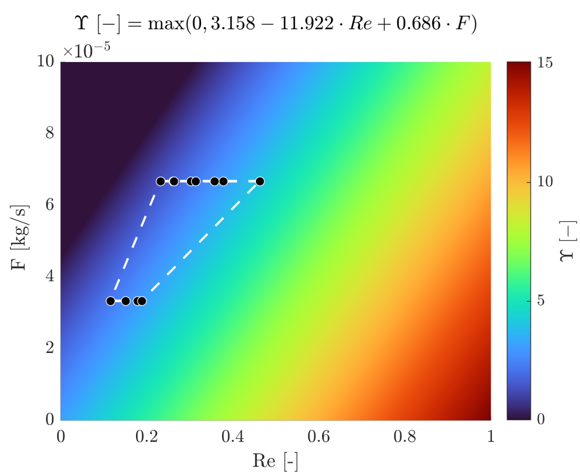

Figure 4: Multiple linear regression $\Upsilon = f(Re, F)$.

same temperature and pressure. As presented by Gmehling et al. [45], the enthalpy departure function for the Peng-Robinson equation of state is defined by Equation 17:

$$h - h^{id} = RT\left[T_r(Z-1) - 2.078(1+\kappa)\sqrt{\alpha(T)}\ln\left(\frac{Z+\left(1+\sqrt{2}\right)B}{Z+\left(1-\sqrt{2}\right)B}\right)\right] \tag{17}$$

where α is defined as $\left(1+\kappa\left(1-\sqrt{T_r}\right)\right)^2$, T_r is the reduced temperature, P_r is the reduced pressure, Z is the compressibility factor, κ is a quadratic function of the acentric factor and B is calculated as $0.07780\frac{P_r}{T_r}$.

Equation 17 requires a reference state, which is assumed to be $T_{ref} = 298.15$ K and $P_{ref} = 1.01325$ bar.

A root finder can be used to find a temperature value that minimises the difference between the value of enthalpy coming from the heat balance and the departure functions. The root-finding procedure is repeated at every time step to find a temperature profile along the spatial direction z.

$$\min_{T}\left(\underbrace{h(t,x)}_{\text{Heat balance}} - \underbrace{h\left(T, P, \rho_f(T,P)\right)}_{\text{Departure function}}\right)^2 \tag{18}$$

2.2.8. *Pressure term*

As explained in Section 2.1, the pressure in the low-velocity region remains nearly constant due to the small pressure wave propagation occurring at the speed of sound. Under such conditions, the term $\partial P/\partial t$ can be approximated by a forward difference equation, describing the pressure change over a small time step Δt. The pressure P in the system is treated as a state variable, while the incoming pressure at the new time step P_{in} is treated as a control:

$$\frac{\partial P}{\partial t} \approx \frac{P_{in} - P}{\Delta t} \tag{19}$$

Such a simplified equation allows for instantaneous pressure. In a real system, the pressure dynamics would depend on the behaviour of the pump and the back-pressure regulator, which introduce additional inertia and resistance to pressure changes, leading to gradual pressure build-up.

2.2.9. *Extraction yield*

The process yield is calculated according to Equation 20, as presented by Sovova et al. [27]. The measurement equation evaluates the solute's mass at the extraction unit outlet and sums it up. The integral form of the measurement (Equation 20) can be transformed into the differential form (Equation 21) and augmented with the process model.

$$y = \int_{t_0}^{t_f} \frac{F}{\rho_f} c_f\Big|_{z=L} dt \tag{20}$$

$$\frac{dy}{dt} = \frac{F}{\rho_f} c_f\Big|_{z=L} \tag{21}$$

The measurement error can be obtained based on the accuracy of the measuring device, or empirically from the spread of data around the model output. The first approach can be utilised only in the absence of external error sources. Given the imperfections of the experimental setup, such as delayed measurements and residuals of the product in pipes and the draining vessel, it was decided that the second approach is more suitable for this work. Based on the Sliczniuk and Oinas [44], the empirical measurement error was estimated to be 0.03.

2.2.10. *Initial and boundary conditions*

It is assumed that the solvent is free of solute at the beginning of the process $c_{f0} = 0$, that all the solid particles have the same initial solute content c_{s0} and that the system is isothermal, hence the initial state is h_0. The fluid at the inlet is considered not to contain any solute. The initial and boundary conditions are defined as follows:

$$\begin{aligned}
&c_f(t=0,z) = 0 \qquad c_s(t=0,z) = c_{s0} \qquad h(t=0,z) = h_0 \\
&c_f(t,z=0) = 0 \qquad h(t,z=0) = h_{in} \qquad \frac{\partial c_f(t,z=L)}{\partial x} = 0 \\
&\frac{\partial h(t,z=L)}{\partial x} = 0 \quad c_s(t,z=\{0,L\}) = 0 \quad y(0) = 0 \quad P(0) = P_0
\end{aligned}$$

2.2.11. Discretisation methods

$$\dot{x} = \frac{dx}{dt} = \begin{bmatrix} \frac{dc_{f,1}}{dt} \\ \vdots \\ \frac{dc_{f,N_z}}{dt} \\ \hline \frac{dc_{s,1}}{dt} \\ \vdots \\ \frac{dc_{s,N_z}}{dt} \\ \hline \frac{dh_1}{dt} \\ \vdots \\ \frac{dh_{N_z}}{dt} \\ \hline \frac{dP}{dt} \\ \hline \frac{dy}{dt} \end{bmatrix} = \underbrace{\begin{bmatrix} G_1\left(c_f, c_s, h; \Theta\right) \\ \vdots \\ G_{N_z}\left(c_f, c_s, h; \Theta\right) \\ \hline G_{N_z+1}\left(c_f, c_s, h; \Theta\right) \\ \vdots \\ G_{2N_z}\left(c_f, c_s, h; \Theta\right) \\ \hline G_{2N_z+1}\left(c_f, c_s, h; \Theta\right) \\ \vdots \\ G_{3N_z}\left(c_f, c_s, h; \Theta\right) \\ \hline G_{3N_z+1}\left(c_f, c_s, h; \Theta\right) \\ \hline G_{3N_z+2}\left(c_f, c_s, h; \Theta\right) \end{bmatrix}}_{G(x;\Theta)}$$

Figure 5: Discretised state space.

The method of lines is used to transform the process model equations into a set of ODEs denoted by $G(x;\Theta)$. For a derivative to be conservative, it must form a telescoping series. In other words, only the boundary terms should remain after adding all terms coming from the discretisation over a grid, and the artificial interior points should be cancelled out. Discretisation is applied to the conservative form of the model to ensure mass conservation. The backward finite difference is used to approximate the first-order derivative, while the central difference scheme approximates the second-order derivative z direction. The length of the fixed bed is divided into N_z, i.e. equally distributed points in the z direction. The state-space model after discretisation is denoted by x and defined as shown in Figure 5, where $x \in \mathbb{R}^{N_x=3N_z+2}$ and $\Theta \in \mathbb{R}^{N_\Theta=N_\theta+N_u}$, N_θ is the number of parameters and N_u is the number of control variables.

3. Variance-based sensitivity analysis

Sobol [50] suggested a probabilistic framework that decomposes the variance of the model's or system's output into fractions that can be attributed to specific inputs or sets of inputs. Variance-based measures of sensitivity are attractive because they measure sensitivity across the entire input space, can handle non-linear responses, and can assess the effect of interactions in non-additive systems. If the process model describes a dynamical system, then it can be said that the model output is

$$Y(t|\theta) = G(x(t|\theta)) = G\left(x\left(t|\theta_1, \theta_2, \cdots, \theta_{n_\theta}\right)\right) \tag{22}$$

where Y is the model output, and θ is a parameter set. Sobol' suggested decomposing the function G into summands of increasing dimensionality:

$$\begin{aligned} G\left(x\left(t|\theta_1, \theta_2, \cdots, \theta_{n_\theta}\right)\right) &= G_0(t) + \sum_{i=1}^{n_\theta} G_i(t, \theta_i) \\ &+ \sum_{i=1}^{n_\theta}\sum_{j=i+1}^{n_\theta} G_{ij}\left(t|\theta_i, \theta_j\right) + \cdots + G_{1\ldots n_\theta}\left(t|\theta_1, \theta_2, \cdots, \theta_{n_\theta}\right) \end{aligned} \tag{23}$$

Suppose the input factors are independent, and each term in this equation is chosen with zero average and is square integrable. In that case, the G_0 is a constant, equal to the expectation value of the output, and the summands are mutually orthogonal. Additionally, this decomposition is unique.

The total unconditional variance can be defined as

$$V(t) = \int_\Omega G^2\left(x(t|\theta)\right) d\theta - G_0(t) \tag{24}$$

with Ω representing the n_θ-dimensional unit hyperspace, where the parameter ranges are scaled from 0 to 1. The partial variances, which are the components of the total variance decomposition, are computed from each of the terms in Equation 22 as

$$V_{i_1,i_2,\cdots,i_s} = \int_0^1 \cdots \int_0^1 G^2_{i_1,\cdots i_s}(t|\theta_{i_1}, \cdots, \theta_{i_s}) d\theta_{i_1} \cdots d\theta_{i_s} \tag{25}$$

where $1 \leq i_1 \leq \cdots \leq i_s \leq p$ and $s = 1, \cdots n_\theta$. With the assumption that the parameters are mutually orthogonal, the variance decomposition becomes

$$V(t) = \sum_{i=1}^{n_\theta} V_i(t) + \sum_{i=1}^{n_\theta-1}\sum_{j=i+1}^{n_\theta} V_{ij}(t) + V_{1,\cdots n_\theta} \tag{26}$$

In this way, the variance contributions to the total output variance of individual parameters and parameter interactions can be determined. These contributions are characterised by the ratio of the partial variance to the total variance, the Sobol's sensitivity indices. The first-order Sobol' index, $S_i(t) = \frac{V_i(t)}{V(t)}$, is a measure for the variance contribution of the individual parameter θ_i to the total model variance. The partial variance $V_i(t)$ is given by the variance of the conditional expectation $V_i(t) = V[\mathbb{E}(Y(t|\theta_i))]$ and is called the "main effect" of θ_i on $Y(t)$. It can be described as the fraction of the model output variance that would disappear on average when θ_i is fixed to a value in its range. The impact on the model output variance of the interaction between θ_i and θ_j is given by the second order Sobol' index, $S_{ij} = \frac{V_{ij}}{V}$, and the total Sobol' index $S_{Ti} = S_i + \sum_{j\neq i} S_{ij}$. They describe it as the result of the main effect of X_i and all its interactions with the other parameters (up to the n_θ'th order).

For additive models and under the assumption of orthogonal input factors, S_{Ti} and S_i are equal, and the sum of all S_i (and thus all S_{Ti}) is 1. For non-additive models, interactions exist, S_{Ti} is greater than S_i, and the sum of all S_i is less than 1. On the other hand, the sum of all S_{Ti} is greater than 1. By analysing the difference between S_{Ti} and S_i, the interactions between parameters can be determined.

To compute the variance and obtain sensitivity measures, Sobol proposed a shortcut calculation based on the assumption that the decomposition's summands are mutually orthogonal. The shortcut is attained by transforming the double-loop integral into an integral of the product of $G(\theta_{j1}, \cdots, \theta_{k-s'}, \theta_1, \cdots, \theta_s)$ and $G(\theta'_{j1}, \cdots, \theta'_{k-s'}, \theta_1, \cdots, \theta_s)$.

Due to the complexity and non-linearities in process models, the analytical solution are difficult to obtain and as such the Monte Carlo integration is often used. The estimation of the total unconditional variance becomes:

$$V(t) \approx \hat{V}(Y(t)) = \frac{1}{n-1}\sum_{m=1}^{n} G^2(\theta_m) - \hat{G}_0^2 \tag{27}$$

$$\hat{G}_0 \approx \frac{1}{n}\sum_{m=1}^{n} G(t|\theta_m) \tag{28}$$

where ˆ stands for the estimate, θ_m is a sampled set of input parameters and n is the number of samples. Saltelli et al. [51] suggested to use two independent input sample $n \times n_\theta$-matrices (the "sample" matrix M_1 and the "resample" matrix M_2) to compute the Monte Carlo integrals, where n is the sample size and n_θ is the number of parameters. Every row in M_1 and M_2 represents a possible parameter combination for the model. The parameter ranges in the matrices are scaled to 0-1. The two matrices are necessary to compile the sample set for $G(\theta_{j1}, \cdots, \theta_{k-s'}, \theta_1, \cdots, \theta_s)$ and $G(\theta'_{j1}, \cdots, \theta'_{k-s'}, \theta_1, \cdots, \theta_s)$.

After generating the two primary matrices, the Saltelli algorithm constructs additional input matrices by replacing columns from the first matrix (base matrix) with columns from the second matrix (resample matrix). The main idea behind the substitution is to isolate the influence of each individual parameter when computing sensitivity indices. The total number of matrices becomes the number of parameters plus two, and the computational complexity is reduced if compared to naive implementations.

3.1. Analysis of the first-order indices

In this section, the GSA is applied to analyse the influence of operating conditions (temperature between 30 and 40 ° C, pressure between 100 and 200 bar and mass flow rate between 3.3 and 6.7 kg/s) on the model output. The main idea behind this analysis is to identify which controls have the largest impact on extraction yield at different times. When evaluating the Sobol indices, a set of output distributions was obtained for every selected time. The average yield curve and confidence regions are shown in Figure 6. It should be noted that the total amount of extract is relatively low (due to lab-scale experiments), so attention should be paid to ensure accurate measurements. The system error might arise due to imprecise control or human error while collecting the samples.

The time evolutions of the total and first-order Sobol sensitivity indices are presented in Figure 7. Initially, the extraction process is strongly dominated by the solvent mass flow rate as the first-order Sobol index for flow rate (S_F) is the most significant contributor to the output variance. As dictated by the mass-transfer limit, the fresh solvent removes the accessible solute from the solid particles, thereby determining the extraction efficiency. Thermodynamic factors (temperature and pressure) have a low impact because the process operates far from equilibrium. This behaviour is evident from the first-order temperature and pressure indices, which initially stay close to zero.

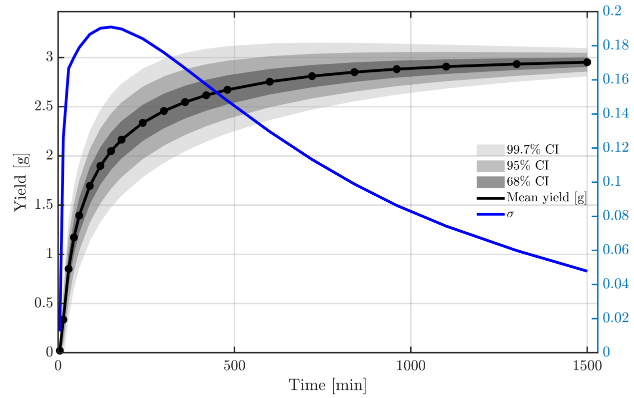

Figure 6: Average yield curve and confidence intervals

As extraction progresses, the importance of operating parameters shifts. Although initially the largest Sobol index is associated with mass flow rate (S_F), it gradually decreases over time (0.31). Over time, mass transfer becomes more dependent on the solvent's capacity to dissolve and transport the solute within the solid particles. When the extraction reaches the equilibrium-limited regime, pressure (S_P) becomes the dominant factor, reaching 0.72 by the final time. The temperature begins to exert a low but measurable influence, and consequently S_T increases to 0.07.

The above results align with classical mass transfer theory and can be interpreted in terms of the flow regimes. The period during which the mass flow rate has the greatest impact corresponds to the kinetic regime. The period during which thermodynamic factors contribute more to the output variance than the flow rate does corresponds to the diffusion regime.

3.2. Analysis of the interaction terms

In Sobol sensitivity analysis, interaction terms quantify how the combined effect of two or more parameters differs from the sum of their individual effects. The presence of interactions indicates that parameters do not act independently. The total Sobol index is defined by Equation 26, and if the interaction term is negligible, then $V(t) \approx \sum_{i=1}^{n_\theta} V_i(t)$. The interaction term can be approximated by subtracting the individual terms from the total Sobol index. Figure 8 summarises the time evolution of the second and third order coefficients.

The negative interaction indices indicate sub-additive variance structure, where the joint variance is less than the sum of individual variances. This means that the joint variance is less than the sum of individual variances:

$$V\left[\mathbb{E}(Y(t|\theta_i, \theta_j))\right] < V\left[\mathbb{E}(Y(t|\theta_i))\right] + V\left[\mathbb{E}(Y(t|\theta_j))\right] \tag{29}$$

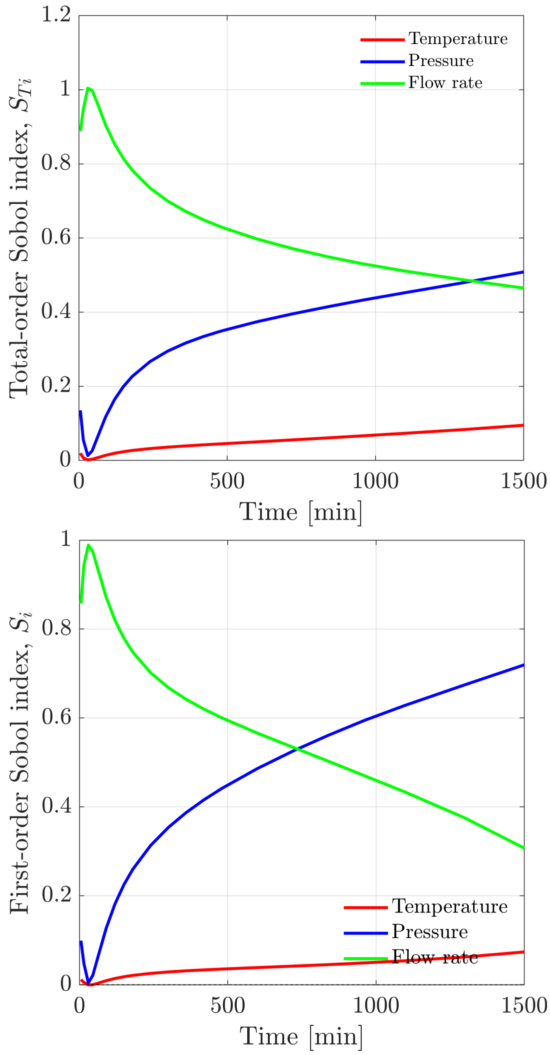

Figure 7: Time evolution of the Sobol indices

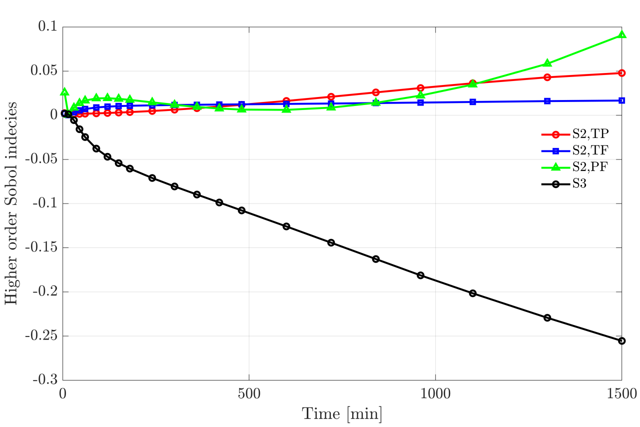

Figure 8: Time evolution of the interaction term

Second-order pairwise interactions remain modest, but increase non-monotonically. These positive interactions arise from the T-P coupling, as reflected in thermodynamic properties (such as CO_2 density) that can take the same values for different combinations of T and P. On the other hand, the P-F interaction can be observed through the Reynolds number. The near-zero temperature-flow interaction (S_{TF}) suggests that this combination of inputs has a low impact on the model output. The temporal evolution of the pressure-flow interaction starts from a low value and shows "wave"-like behaviour. After the first positive extremum, S_{PF} flattens and then increases exponentially.

The opposing effects of temperature and pressure on CO_2 properties, such as density, should be noted. The temperature decreases density ($\partial\rho/\partial T < 0$) while pressure increases it ($\partial\rho/\partial P > 0$). If the variations in both inputs are considered together, these effects partially cancel each other out. This leads to a smaller range of density values than the sum of the individual contributions. Such redundancy suggests that the second-order interactions for T and P should be negative, but model non-linearities should also be considered. The non-linearity of the physical properties around the critical point is profound, which leads to variance increase in that region. The non-zero cross-derivative ($(\partial^2\rho/(\partial P\partial T))$) introduces additional variance that cannot be obtained from knowing T or P separately.

The third-order interaction index takes negative values (reaching $S_3 = -0.26$ at $t = 1500$ min), indicating that the resulting output variance is lower than predicted by summing individual effects and pairwise interactions when all three parameters vary simultaneously. This is justified by parameter redundancy, which is larger than in the second-order case. The redundancy from the three-way interaction exceeds the pairwise non-linearities, which reduce the output variance less than the redundancy, resulting in negative values of S_3.

One practical implication of first-order response surface methodology and first-order design of experiments is that they neglect higher-order interactions, which can lead to overestimation of process output. Another observation coming from the Sobol analysis is the time evolution in parameter importance. The results show a shift from flow-dominated to pressure-dominated. Consequently, adaptive control strategies that adjust parameter priorities across extraction phases are suggested. The appropriate strategy would focus on flow rate during the kinetic-limited

4. Conclusion

This work focus on application and comparison of variance-based diagnostic to measures an impact of controls on a SFE model. The analysis of results helped to identify which of controls is the most influential. Later the analysis was extended to investigate how a combination of controls affect the system. Due to dynamic nature of the model, a temporal evolution of the Sobol indices was observed. The results were analysed not only with respect to the future application of the dynamic optimization but also from the perspective of mass transfer theory.

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